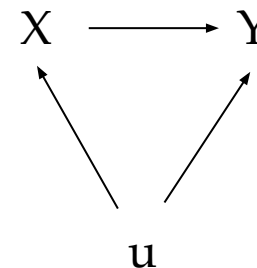

Intro to Instrumental Variables

- We want to estimate the *causal* effect of X on Y
- $Y = \beta X + u$
 - Where u is the error term
- Let's say we are in a perfect world (or close to, e.g. we randomize treatment). Then β would capture the causal effect of X on Y
 - How much Y changes, on average, for a unit change in X
- We are hardly in a perfect world though

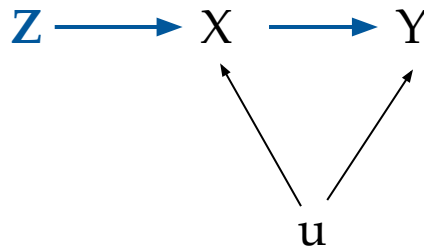
Intro to Instrumental Variables

- When the regressor X is not exogenous, i.e. $\text{Corr}(X, u) \neq 0$...
 - Shown in the figure by the arrow connecting u to X
- ... we cannot interpret the relationship between X and Y as causal
- What can we do?
- That's where IV can be helpful



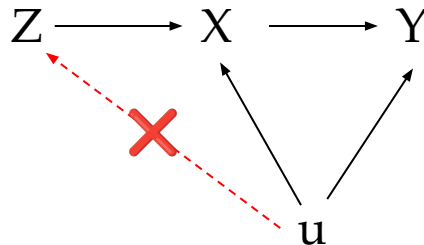
Intro to Instrumental Variables

- An IV (let's call it **Z**) is a variable that as it varies...
- ...causes our endogenous regressor (X) to vary
- ...in a way that is uncorrelated with the error term
- ...and let that “clean” variation in X measure the causal effect
- **That's it!**



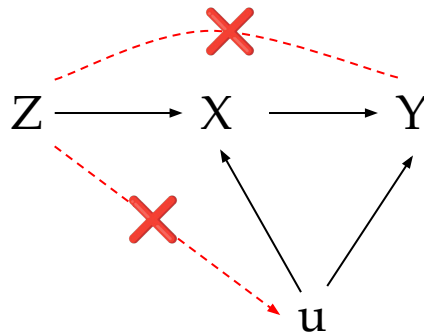
Properties of IVs

- **Instrument exogeneity**: The instrument is uncorrelated with the error term, i.e. $\text{Corr}(Z, u) = 0$



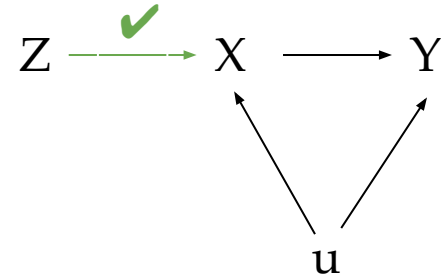
Properties of IVs

- Instrument exogeneity also implies that the instruments themselves do not belong in the regression (so called **Exclusion restriction**)
 - The only way the excluded exogenous variable (the IV) can influence the dependent variable (Y) is through its influence on the regressor (X)



Properties of IVs

- **Instrument relevance:** The instrument is correlated with the regressors of the model (including of course X), i.e., in this case: $\text{Corr}(Z, X) \neq 0$



In sum, a good IV:

- Should be a good predictor of X (correlated with X)
- Should be exogenous (uncorrelated with u)
- Should affect Y only through X

Example: Ciccone and Hall (1996)

- Estimating equation (simplified): $Y = \beta X + u$
- Y = Productivity
- X = Density
- *If there are true differences in the determinants of productivity across states, it is not realistic to assume that density is uncorrelated with those differences. (p.60)*
 - i.e., $\text{Corr}(X, u) \neq 0$

Example: Ciccone and Hall (1996)

- Their proposed instruments: Population in 1850, Population density in 1880, Presence/absence of railroad in 1860, Distance from eastern seaboard
 - All proxies for past agglomeration patterns
- Let's think: Are the IVs (1) relevant, (2) exogenous, and (3) do they satisfy the exclusion restriction?
- **Relevance:** Arguably so (we may well expect that past population correlates with current population)
 - The instrument relevance is also something that can be checked with the data (graphically, or better with regression of X on Z)

Example: Ciccone and Hall (1996)

- Exogeneity/Exclusion restriction
- The author's argument (p.61): *Our hypothesis is that the early patterns of agglomeration in the US did not reflect factors which significantly contribute to productivity today...*
 - i.e. $\text{Corr}(Z, Y) = 0$
- *...but have a remaining influence mainly through the legacy of agglomeration.*
 - i.e. $Z \rightarrow X \rightarrow Y$
 - which as we have seen, means: $\text{Corr}(Z, u) = 0$ & $\text{Corr}(Z, Y) = 0$

Example: Ciccone and Hall (1996)

- The exclusion restriction could be questioned: there are potentially some other sources of persistence (besides the legacy of agglomeration) that would enter the error term if not accounted (controlled) for
- For instance, more accumulation of human capital in historically more populated places; local institutions,...

Example: Ciccone and Hall (1996)

